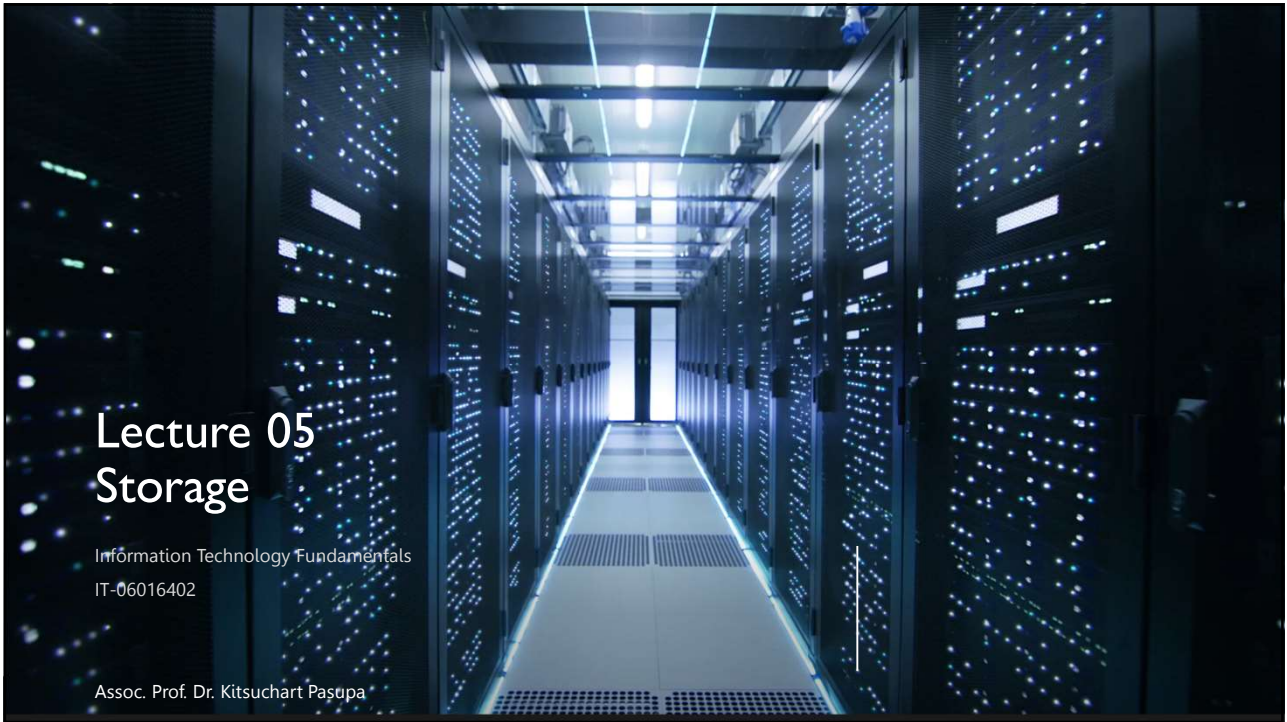




1

Storage

Primary Storage	Secondary Storage	Offline Storage	Tertiary Storage
• It is the main memory where data and information are stored • It is called “memory” • Volatile/Non Volatile in nature • Examples are RAM, ROM, etc.	• It is referred to the external memory where data is stored <u>permanently</u> • Long-time storage • It is called “storage medium” • Non-volatile in nature • Examples are HDD, SSD, etc.	• It is called disconnected or removable storage. • It is a computer data storage on a medium or a device that is not under the control of a processing unit • It must be inserted or connected by a human operator before a computer can access it again. • Examples are CD DVD USB Flash Drive	• Similar to secondary storage, but can only be accessed via a network device. • Access is even slower. • It uses additional robotic mechanisms to transfer media between their long-term storage locations and available drives without human intervention. • Examples are NAS SAN – Tape Optical Disc

2

Storage

A storage medium is the physical material on which a computer keeps data, information, programs, and applications

Cloud storage keeps information on servers on the Internet, and the actual media on which the files are stored are transparent to the user

3

Storage Device

- A storage device is the hardware that records and/or retrieves items to and from storage media



Read

• Reading is the process of transferring items from a storage medium into memory

Input



Write

• Writing is the process of transferring items from memory to a storage medium

Output

It is not I/O device

4

Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



Memory Cards



USB Flash Drive



Cloud Storage



Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags



Film

5

Capacity

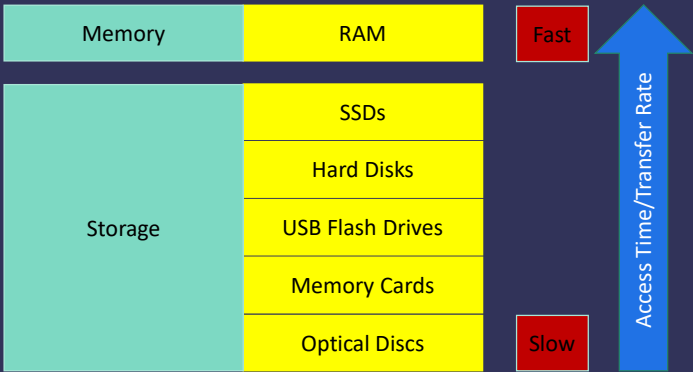
- Capacity is the number of bytes a storage medium can hold

Storage Term	Short Form	Approximate Number of Byte	Exact Number of Byte
Kilobyte	KB	1 thousand (10 ³)	2 ¹⁰ = 1,024
Megabyte	MB	1 million (10 ⁶)	2 ²⁰ = 1,048,576
Gigabyte	GB	1 billion (10 ⁹)	2 ³⁰ = 1,073,741,824
Terabyte	TB	1 trillion (10 ¹²)	2 ⁴⁰
Petabyte	PB	1 quadrillion (10 ¹⁵)	2 ⁵⁰
Exabyte	EB	1 quintillion (10 ¹⁸)	2 ⁶⁰
Zettabyte	ZB	1 sextillion (10 ²¹)	2 ⁷⁰
Yottabyte	YB	1 septillion (10 ²⁴)	2 ⁸⁰

6

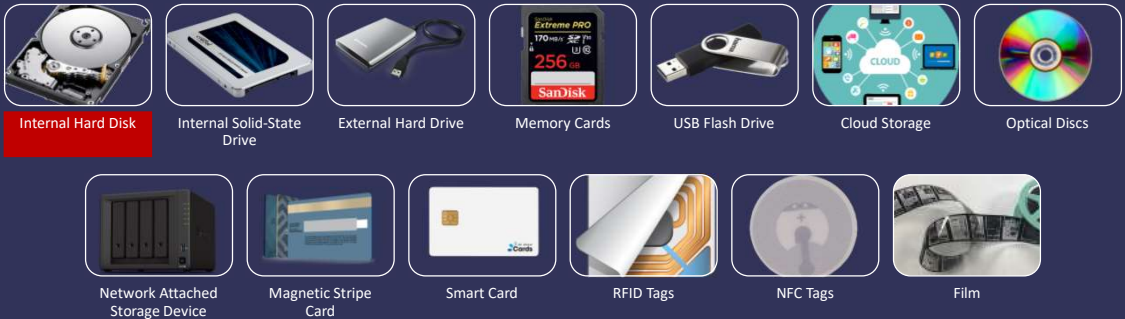
Access Time

- Access time measures:
 - The amount of time it takes a storage device to locate an item on a storage medium
 - The time required to deliver an item from memory to the processor
- Some manufacturers use transfer rate
 - Kbps (kilobyte per second)
 - Mbps (Megabyte per second)
 - Gbps (Gigabyte for second)



7

Storage



8

Hard Disk

- A hard disk, also called a hard disk drive (HDD) contains one or more inflexible, circular platters that use magnetic particles to store data, instructions, and information



9

Hard Disk

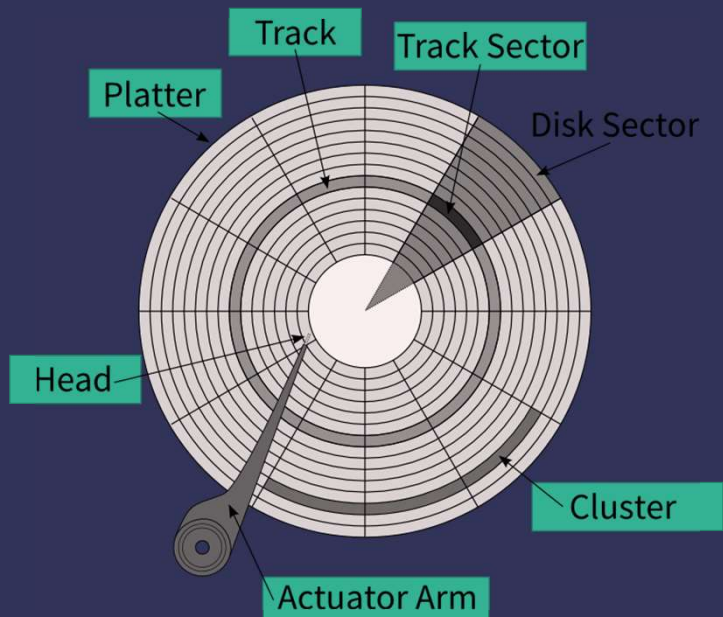
- The storage capacity of hard disks varies and is determined by:
 - The number of platters the hard disk contains
 - Whether the disk uses longitudinal or perpendicular recording
 - Density



10

Hard Disk

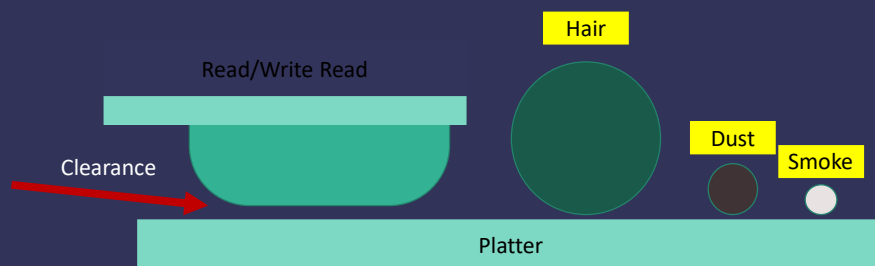
- The two most common form factors for modern HDDs are 3.5-inch, for desktop computers, and 2.5-inch, primarily for laptops
- Disk read/write heads are the small parts of a disk drive which move above the disk platter and transform the platter's magnetic field into electrical current (read the disk) or, vice versa, transform electrical current into magnetic field (write the disk).
- Formatting is the process of dividing the disk into tracks and sectors
- Revolutions per minute, RPM, is used to help determine the access time on computer hard drives. Most common RPM rates, in both laptop and desktop PCs, are between 5400 and 7200 RPM.



11

Hard Disk

- A head crash occurs when a read/write head touches the surface of a platter
- Always keep a backup of your hard disk



12

Hard Disk

Mechanical parts of hard disk directly affect the access time

Disk cache, sometimes called a buffer, consists of a memory chip(s) on a hard disk that stores frequently accessed data, instructions, and information

The larger the disk cache, the faster the hard disk

13

Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



Memory Cards



USB Flash Drive



Cloud Storage



Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags



Film

14

Solid-state Drive

- SSD is a flash memory storage device that contains its own processor to manage its storage
- An SSD has several advantages over traditional (magnetic) hard disks:

Faster access times	Faster transfer rates	Quieter operation
More durable	Lighter weight	Less power consumption
Less heat generation	Longer life	Defragmentation not required



15

Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



Memory Cards



USB Flash Drive



Cloud Storage



Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags



Film

16

External Hard Disk

- An external hard disk is a separate freestanding storage device that connects with a cable to a USB port or other port on a computer or mobile device



17

Redundant Array of Independent Disks

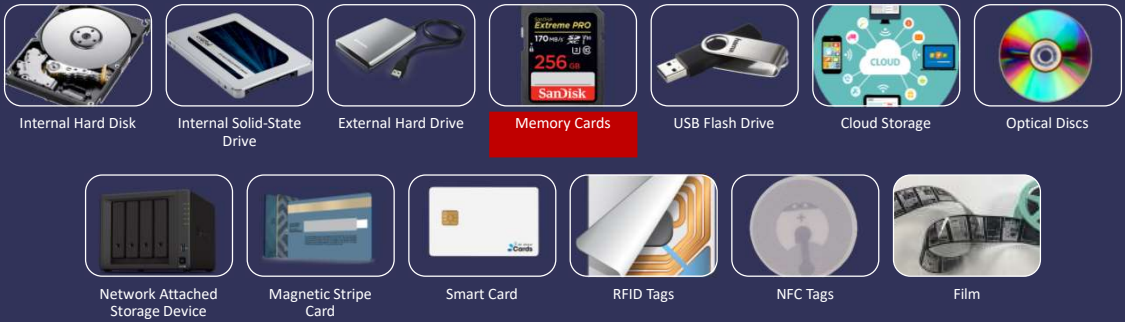
- RAID is a data storage virtualization technology
- It is a group of two or more integrated hard disks or SSDs that is for the purposes of data redundancy, performance improvement, or both.



<https://en.wikipedia.org/wiki/RAID>

18

Storage

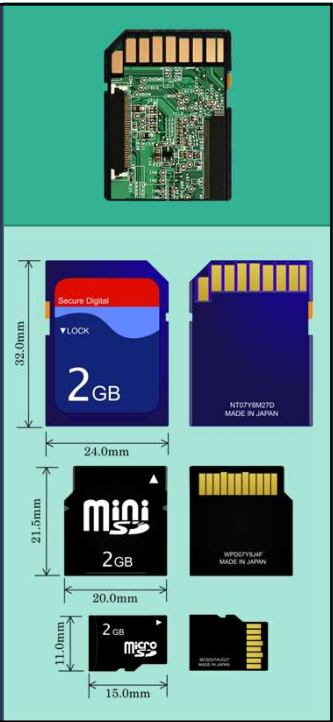


19

Memory Card

- It is a removable flash memory storage device that you insert and remove from a slot in a computer, mobile device, or card reader/writer

	Secured Digital (SD) Card	2 GB (Max)
	Secured Digital High Capacity (SDHC) Card	32 GB (Max)
	Secure Digital Extended Capacity (SDXC) card	2 TB (Max)
	Secure Digital Ultra Capacity (SDUC) card	128 TB (Max)



20

Memory Card

Minimum sequential write speed	Speed Class			Corresponding video format			
	Speed Class	UHS Speed Class	Video Speed Class (new)	8K video	4K video	Full HD / HD video	Standard Video
Card image				The necessary speed varies by each recording/playback device condition, even in the same format.			
90MB/sec			V90	✓			
60MB/sec			V60	✓	✓		
30MB/sec		U	V30	✓	✓	✓	
10MB/sec	@	U	V10		✓	✓	✓
6MB/sec	@		V6		✓	✓	✓
4MB/sec	@					✓	✓
2MB/sec	@						✓

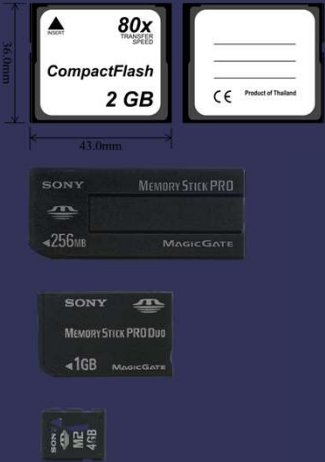
<https://www.kingston.com/en/solutions/personal-storage/memory-card-speed-classes>

21

Memory Card

Compact Flash

Memory Stick



22

Memory Card



Card Reader/Writer



Printer



Laptop



Smartphone/Tablet

23

Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



Memory Cards



USB Flash Drive



Cloud Storage



Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags

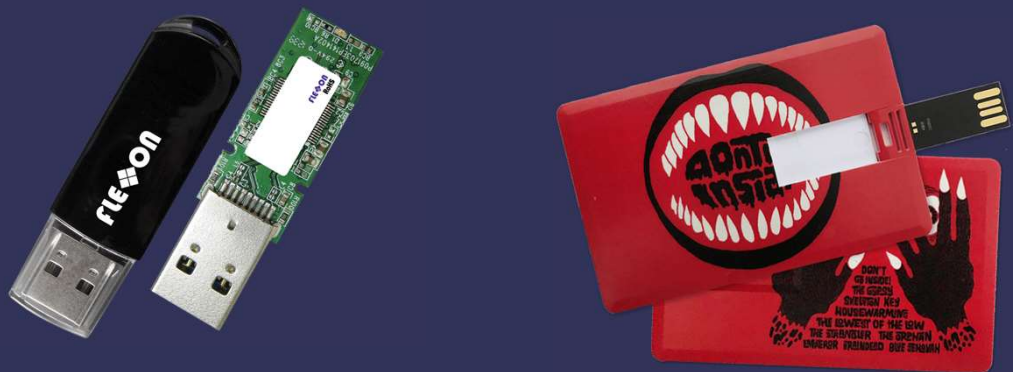


Film

24

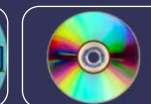
USB Flash Drive

- USB flash drives plug into a USB port on a computer or mobile device



25

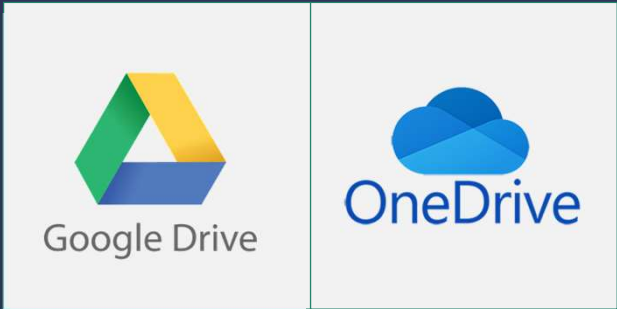
Storage

 Internal Hard Disk	 Internal Solid-State Drive	 External Hard Drive	 Memory Cards	 USB Flash Drive	 Cloud Storage	 Optical Discs
 Network Attached Storage Device	 Magnetic Stripe Card	 Smart Card	 RFID Tags	 NFC Tags	 Film	

26

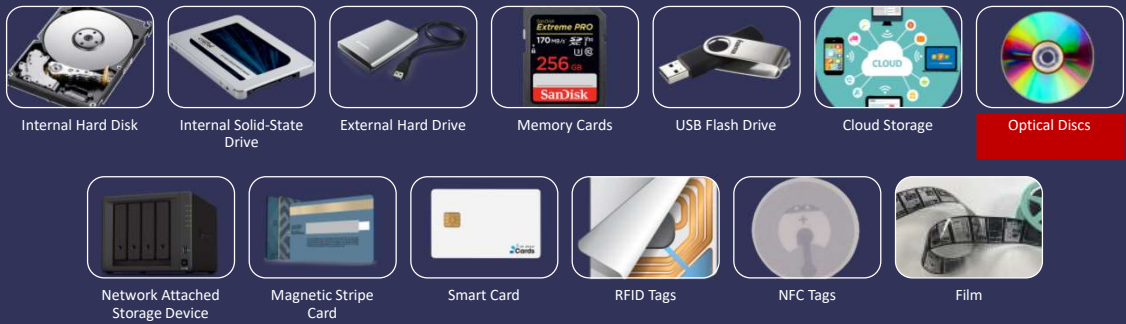
Cloud Storage

- Cloud storage is an Internet service that provides storage to computer or mobile device users



27

Storage



28

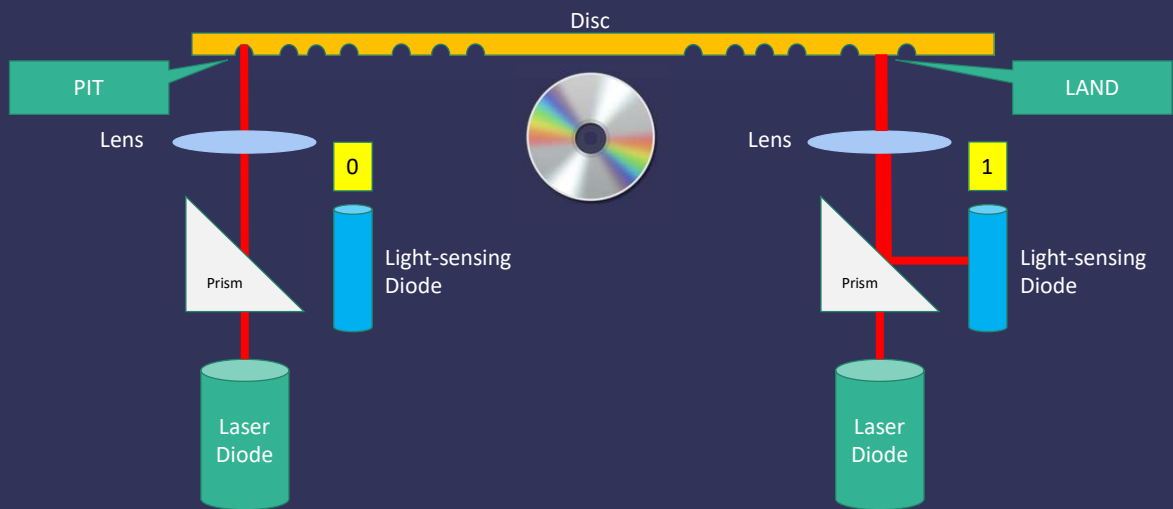
Optical Disc

- An optical disc consists of a flat, round, portable disc made of metal, plastic, and lacquer that is written and read by a laser



29

How a Laser Reads Data on an Optical Disc



30

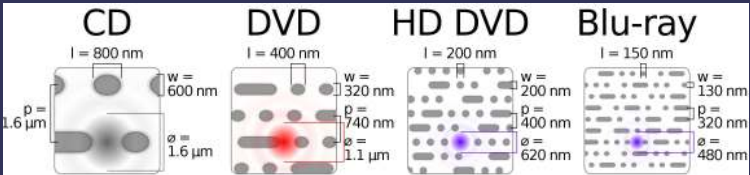
CD –vs– DVD



CD



DVD

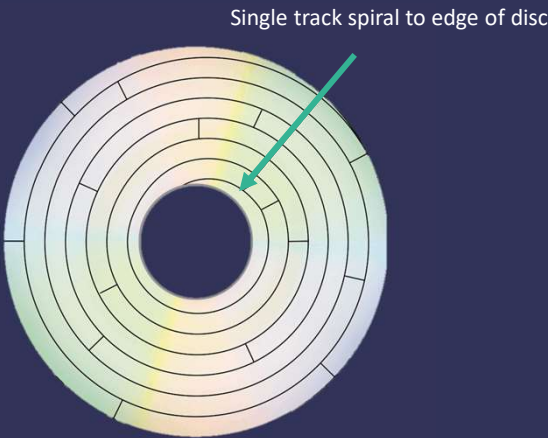


https://en.wikipedia.org/wiki/Optical_disc

31

Optical Disc

- Optical discs commonly store items in a single track that spirals from the center of the disc to the edge of the disc
- Track is divided into evenly sized sectors



32

CD –vs– DVD

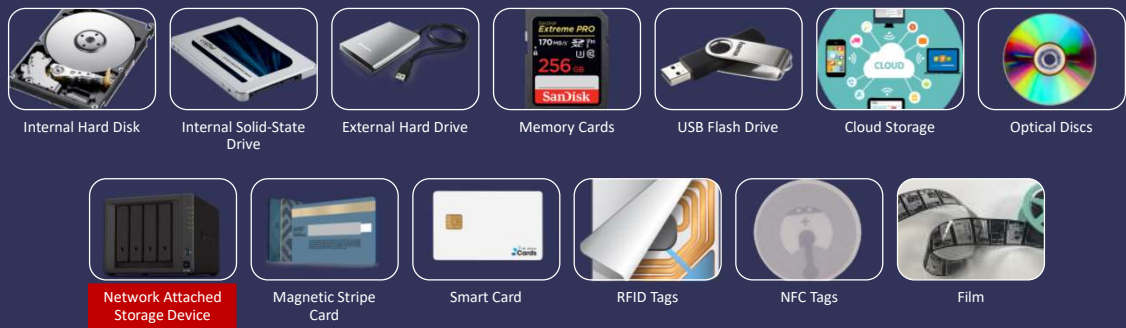
- CD

 - A CD-ROM can be read from but not written to
 - Single-session disc
 - A CD-R is an optical disc on which users can write once, but not erase
 - A CD-RW is an erasable multisession disc
- DVD

 - A DVD-ROM is a high-capacity optical disc on which users can read but not write on or erase
 - A DVD-R or DVD+R are competing DVD-recordable WORM formats, on which users can write once but not erase
 - DVD-RW, DVD+RW, and DVD+RAM are competing DVD-rewritable formats that users can write on multiple times

33

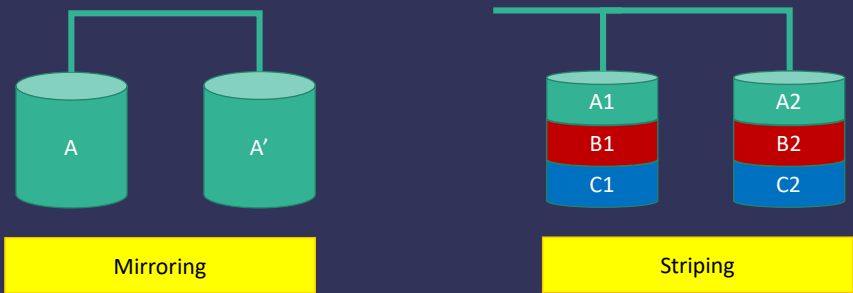
Storage



34

Enterprise Storage

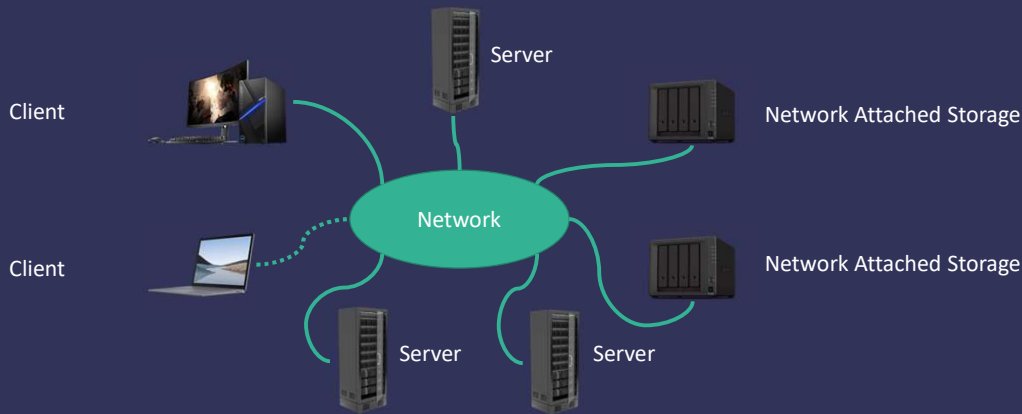
- Enterprise hardware allows large organizations to manage and store data and information using devices intended for heavy use, maximum efficiency, and maximum availability
- It employs RAID to duplicate data, instructions, and information to improve data reliability



35

Network attached storage (NAS)

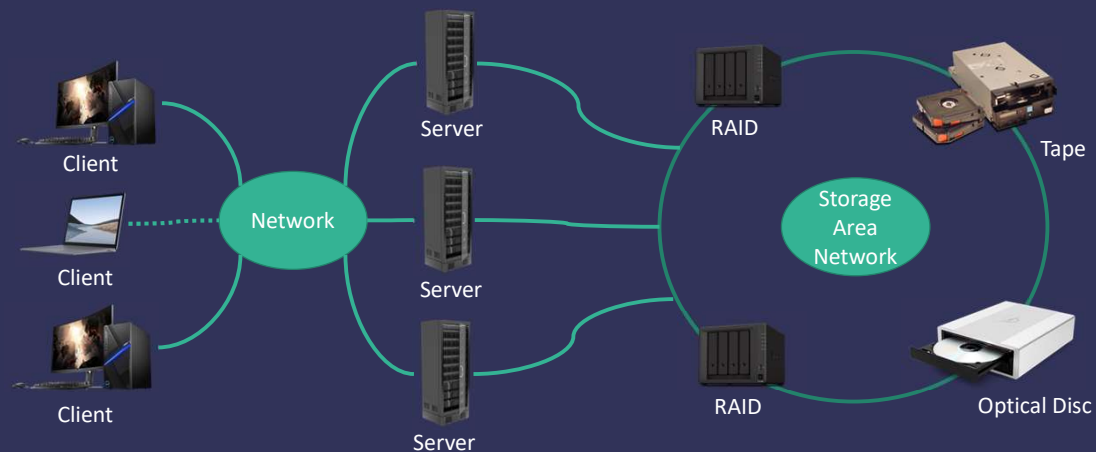
- NAS is a server that is placed on a network with the sole purpose of providing storage to users, computers, and devices attached to the network



36

Storage Area Network (SAN)

- A storage area network (SAN) is a high-speed network with the sole purpose of providing storage to other attached servers



37

Magnetic Tape System

- Tape is a magnetically coated ribbon of plastic capable of storing large amounts of data and information
- A tape drive reads and writes data and information on a magnetic tape



38

Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



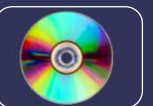
Memory Cards



USB Flash Drive



Cloud Storage



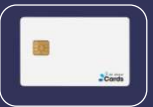
Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags



Film

39

Other Types of Storage



Magnetic Stripe Card



Smart Card



RFID Tag

40

Near Field Communications

An NFC-enabled device contains an NFC chip

An NFC tag contains a chip and an antenna that contains information to be transmitted

Most NFC tags are self-adhesive



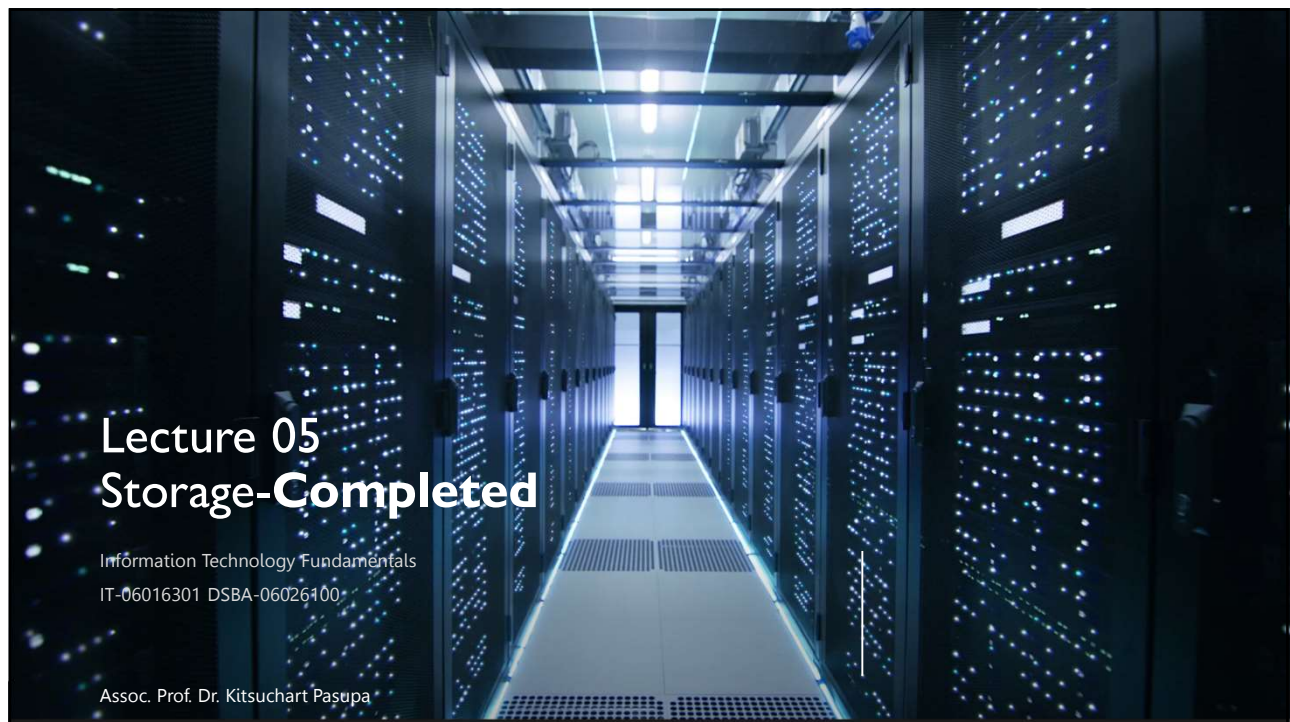
41

Film

- Microfilm and microfiche store microscopic images of documents on a roll or sheet film



42



43